RAKSHITH MATHAD

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EDUCATION

University of Southern California, Los Angeles, California - Master of Science in Applied Data Science AUG 2022-MAY 2024

Coursework Highlights: Machine Learning for Data Science and AI, Applications of Data Mining, Predictive Analytics, Fairness, Security and Privacy in AI

KLE Technological University, India - Bachelor of Engineering in Computer Science AUG 2018-JUN 2022

Coursework Highlights: Data Structures and Algorithms, Data Mining, Computer Architecture, Operating Systems, Distributed & HPC, Cloud Computing

SKILLS

Languages	Python, SQL, C, C++, CUDA (Intermediate), OpenMP (Intermediate)
Tools & Technologies	Analytics, Deep Learning, Machine Learning, A/B Testing, GCP Vertex AI, AWS, Docker, Git, Informatica ETL, Web Scraping/Automation, Parallel Programming, Numpy, Pandas, Scikit learn, LangChain, Nvidia DGX Server, Nsight Profiling, Nvidia NeMo, LLMOps, RAG, VectorDBs, LLM Training and Inference techniques, HuggingFace, FastAPI, RESTful API, Linux, MongoDB, Selenium, Hadoop HDFS, PySpark, PowerBI/Tableau, LLMs, Generative AI, Azure, Google Cloud Platform, BigQuery, Kubernetes**,** HiveQL, Horovod, ArcGIS, Jenkins CI/CD, NLP, Forecasting, Unsupervised ML
Certifications	NVIDIA CUDA Computing ,Juniper Networks Certified Associate, TensorFlow/PyTorch for Artificial Intelligence, Agile, TCP/IP

EXPERIENCE

CVS Health - Software Engineer - Generative AI, New York City, NY JUN 2024- PRESENT

- As a **generalist AI Engineer,** I am working for **Aetna Insurance** in the **Conversational AI** call center customer service team to build a large-scale high volume low latency **RAG** and Rule based **FastAPI** chat application on **GCP Google AI Platform**. My contributions to the project has an **impact of ~ $12 MM/yr and I received a company-wide award for high and vibrancy impact**. Worked with **OpenAI GPT PTUs and Gemini LLMs,** employed **Feature Store** and Matching Engine to build multithreaded/parallel semantic search based information retrieval and answer generation modules, resulting in improved latency and caching strategy. Helped in profiling and **scaling this production AI system** to **~20,000 req/hr impacting around 200k customers everyday** leveraging **Kubernetes GKE** and **Vertex AI**.
- Performed complex data ingestion and preprocessing on GCP **BigQuery** and **BigTable**, including chunking and embedding strategies. Took part in design reviews, building chat **Agents**, improving semantic search by **fine-tuning** the embedding model service with bi-encoders etc.
- Also worked with Airflow, **K8s**, Jenkins CI/CD**, LLM Evaluation and regression framework**, scraping/automation, feature flags, On call rotations with prod deployments, advanced prompt engineering, load testing, AI safety, and guardrails etc. I constantly produced multiple impactful PRs every week, exceeding expectations.

AEG Entertainment Group - Data Engineer Intern, Los Angeles, CA FEB 2024-MAY 2024

- Built complex **Azure** Data Factory pipelines for large data processing, used **PySpark** in **Databricks** for batch API processing and Dynamics CRM data migration via Python OData REST APIs. Handled Parquet/Avro for streaming and financial data reporting.

CVS Health - Analytics Engineering Intern, New York City, NY MAY 2023- AUG 2023

- Optimized Hadoop HDFS big data pipelines, processing **100M+** rows with HiveQL, cutting latency by 40%. Designed scalable ETL workflows and **boosted customer campaign growth** strategy leveraging **Tableau**, advanced OLAP, data blending, and predictive modeling.

AlphaICs Corporation - AI Software Engineer Intern, India JAN 2022-JUL 2022

- Developed deep learning applications for Object Detection, Recurrent Models for Visual Attention, Recommendation Systems (Matrix Factorization, Collaborative Filtering) etc for faster AI **Inference** on a custom **AI processor**.

Samsung R&D Institute -AI Research Intern, India NOV 2020-JUN 2021

- Led a team of 3, and worked on the SOTA **Generative AI PyTorch** research Implementations for the task **of Semantic Image Editing using Conditional GAN** with spatially adaptive normalization, for Image manipulation with controlled generative-fill. Performed Semantic Segmentation with DeepLab-V2 to curate a 20k-image dataset and trained generative models on an **Nvidia DGX** cluster.

PROJECTS

STUDY AND IMPLEMENTATIONS OF PARALLELISM FOR LLM INFERENCE ON GPUs

- Used **Ray Serve** to deploy a async **TinyLlama** on a local GPU using FastAPI and stress test it.
- Implemented a simple hybrid **OpenMP** + **CUDA** vector addition achieving **8.8x** speedup by combining 4 OpenMP threads (CPU parallelism) with individual CUDA streams (GPU parallelism) for overlapped memory transfers and concurrent kernel execution.
- Built a **Retrieval-Augmented Generation (RAG)** pipeline with **LLaMA 2**, integrating **LangChain**, **Pinecone VectorDB**, and **HuggingFace** Sentence Transformers, enhanced with **NVIDIA NeMo Guardrails** and LoRA-based fine-tuning for safer and more efficient responses.

PERFORMANCE PROFILING AND GPU OPTIMIZATION OF BERT INFERENCE PIPELINE

- Deployed and served BERT Transformer on an RTX GPU in **PyTorch, ONNX, NVIDIA TensorRT** runtimes and profiled it to study performance bottlenecks using **NVIDIA Nsight**. Building a small dummy CUDA Kernel to understand infusion into Pytorch for potential optimized inference.

RESEARCH EXPERIENCE

- **Published** a paper "*Performance Analysis of Distributed Deep Learning using Horovod on Image Classification*", on **Horovod ring all reduce** for **multi GPU distributed training** on Nvidia DGX cluster at IEEE **ICICCS** **(Google Scholar)**
- **Machine Learning Engineer (Information Sciences Institue (ISI) – Los Angeles, CA)** – Developed algorithms to leverage ML privacy-preserving techniques to detect anomalies infused by adversary client nodes in a **Federated Learning** system for **Homomorphic Encryption.**